

DATA MANAGEMENT

The Data Management Track is a comprehensive program designed to develop highly skilled data professionals capable of transforming data into strategic business value. Combining Data Engineering, Business Intelligence, Data Warehousing, Big Data, SQL, Data Integration, and Data Governance, the track builds a strong foundation across the technical and business dimensions of modern data ecosystems. Through hands-on experience with industry-leading technologies and best practices, trainees gain the skills needed to design, build, and manage end-to-end data solutions preparing them for impactful careers.

JOB PROFILES

**Data Engineer · Big Data Engineer ·
Business Intelligence Developer / Analyst ·
Data Integration Specialist · Data Warehouse Modeler**



TOTAL HOURS

1,180^{hrs}

Theory + hands-on labs



DELIVERY

Blended

Online & on-site



UNIVERSITY GRADE

Good

Minimum grade required



CERTIFICATIONS

8


Where graduates work

VOIS Vodafone PwC Deloitte CIB BBI Fawry etax
EFG Hermes Etisalat Misr AAIB Fawry CyShield IBM
Giza Systems Valeo



Certifications earned

- AWS Certified Data Engineer – Associate
- IBM Infosphere DataStage 11.5
- Databricks Certified Data Engineer Associate
- DP-900** — Microsoft Certified: Azure Data Fundamentals
- DP-600** — Microsoft Certified: Fabric Analytics Engineer Associate
- PL-300** — Microsoft Certified: Power BI Data Analyst Associate
- DP-700** — Microsoft Certified: Fabric Data Engineer Associate
- DP-750** — Microsoft Certified: Azure Databricks Data Engineer Associate



Selection process

01

IQ & English Exam



Online exam covering problem-solving aptitude and English proficiency.

02

Technical Exam



Computer-based assessment covering the program's core technical fundamentals.

03

Technical Interview



Interview with the technical committee.

04

Interpersonal Interview



Candidates who pass the technical interview are invited to this final stage.



Who may apply

Open to all university graduates with a minimum grade of Good or higher.

Program content – 1,180 hours

CATEGORY	HOURS	SHARE	KEY COURSES
● Fundamental Courses	72 hrs	6%	Computer Network Fundamentals, Database Fundamentals, Operating Systems Fundamentals, Introduction to programming
● Core Courses	751 hrs	64%	Advanced Java SE for Data Management, Automation and CI/CD Essentials, Analytical SQL, Apache Airflow, Apache Hadoop Essentials, Apache HBase Essentials, Apache Kafka, Apache Pig, Hive and Impala, Apache Scala Programming Language, Apache Spark Essentials, Bash Shell Script, Big Data Case Study, Business Objects Webi / Universe, Cloud Computing Fundamentals for Data Analysis, Containerization using Docker, Data Analysis for Data Management, Data Management With Python, Data Visualization using Power BI, Data Warehousing Fundamentals And Data Modeling, Financial Accounting, IBM InfoSphere DataStage Essentials, Incorta Business Analytics, Informatica PowerCenter Basics, Introduction to Oracle SQL and PL/SQL, NOSQL using MongoDB, Object-Oriented Paradigm using Java, Oracle Advanced PL/SQL, Red Hat System Administration I, Red Hat System Administration II, SQL Tuning, Visualization and Storytelling using Tableau, Statistics, Agile
● Graduation Project	300 hrs	25%	The ITI Graduation Project is an intensive, end-to-end capstone where trainees collaborate with industry partners to solve real-world business challenges by designing and deploying production-ready data pipelines, cloud-native architectures, and predictive analytics models. It serves as a comprehensive simulation of enterprise data engineering and business intelligence lifecycles, proving the trainees' readiness for immediate market deployment.
● Soft Skills	57 hrs	5%	Communication Essentials for Professionals, High Impact Presentations, Job Seeking Skills, Professional Demeanor

75% face to face Learning| 25% Online

Hardware requirement: Core i7 laptop with at least 16 GB of RAM

Student Deliverable

Each student must deliver at least ONE freelancing job and an international certificate based on his track

Before you apply

Applicants preferred to have adequate knowledge in Database Fundamentals and Basic Knowledge about technical concepts

Database Fundamentals
<https://maharatech.gov.eg/course/view.php?id=740>

Career Talk in Data Analytics
<https://maharatech.gov.eg/course/view.php?id=514>

Good Fundamentals for Data Warehousing, Big Data, Data Engineering and understanding the Data Analytics life cycle

ITI values that could be found here
<https://iti.gov.eg/about-us>

APPLY AT

ITI.GOV.EG

Applications open each intake period. Check the website for current dates and deadlines.